$$\lambda_5 \to \frac{1}{16\,\pi^2}\,\Big(m_1\,\tilde{\mu}\,g_1^4\,\left(-\overline{C_{\phi2\phi1}}+m_1\,\tilde{\mu}\right)\,\mathrm{LF}_{2,2,0}[m_1,\tilde{\mu}]\,+$$

$$\frac{1}{3}\,m_2\,\tilde{\mu}\,g_2^4\,\left(-5\,\overline{C_{\phi2\phi1}}+9\,m_2\,\tilde{\mu}\right)\,\mathrm{LF}_{2,2,0}[m_2,\tilde{\mu}]-\frac{2}{3}\,m_2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^4\,\mathrm{LF}_{3,2,-1}[m_2,\tilde{\mu}]\,+$$

$$\frac{1}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^2\,\overline{y_e}^{pr}\,a_e^{pr}\,\mathrm{LF}_{4,1,-1}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}\right]-\frac{2}{3}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^2\,\overline{y_e}^{pr}\,a_e^{pr}\,\mathrm{LF}_{5,1,-2}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}\right]+$$

$$\frac{3}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^2\,\overline{y_d}^{pr}\,a_d^{pr}\,\mathrm{LF}_{4,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}\right]-2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^2\,\overline{y_d}^{pr}\,a_d^{pr}\,\mathrm{LF}_{5,1,-2}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}\right]+$$

$$\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^2\,\overline{y_u}^{pr}\,a_u^{pr}\,\mathrm{LF}_{2,2,0}\left[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}\right]-\frac{1}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^2\,\overline{y_u}^{pr}\,a_u^{pr}\,\mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}\right]-$$

$$\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^2\,\overline{y_u}^{pr}\,a_u^{pr}\,\mathrm{LF}_{3,1,0}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}\right]-\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^2\,\overline{y_u}^{pr}\,a_u^{pr}\,\mathrm{LF}_{3,2,-1}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}\right]+$$

$$3\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^2\,\overline{y_u}^{pr}\,a_u^{pr}\,\mathrm{LF}_{4,1,-1}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}\right]-2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^2\,\overline{y_u}^{pr}\,a_u^{pr}\,\mathrm{LF}_{5,1,-2}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}\right]-$$

$$\frac{5}{2}\,m_1\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_1^4\,\mathrm{LF}_{3,2,-1}[\tilde{\mu},m_1]+g_1^4\,m_1^2\,\tilde{\mu}^2\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\,\mathrm{LF}_{3,2,0}[\tilde{\mu},m_1]\,+$$

$$\frac{1}{2}\,m_1\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_1^2\,g_2^2\,\mathrm{LF}_{4,1,-1}[\tilde{\mu},m_1]+2\,m_1\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_1^4\,\mathrm{LF}_{4,2,-2}[\tilde{\mu},m_1]\,-$$

$$g_1^4\,m_1^2\,\tilde{\mu}^2\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\,\mathrm{LF}_{4,2,-1}[\tilde{\mu},m_1]-\frac{2}{3}\,m_1\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_1^2\,g_2^2\,\mathrm{LF}_{5,1,-2}[\tilde{\mu},m_1]\,-$$

$$\frac{4}{3}\,m_2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^4\,\mathrm{LF}_{3,1,0}[\tilde{\mu},m_2]-\frac{59}{6}\,m_2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^4\,\mathrm{LF}_{3,2,-1}[\tilde{\mu},m_2]\,+$$

$$3\,g_2^4\,m_2^2\,\tilde{\mu}^2\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\,\mathrm{LF}_{3,2,0}[\tilde{\mu},m_2]+\frac{7}{2}\,m_2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^4\,\mathrm{LF}_{4,1,-1}[\tilde{\mu},m_2]\,+$$

$$8\,m_2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^4\,\mathrm{LF}_{4,2,-2}[\tilde{\mu},m_2]-4\,g_2^4\,m_2^2\,\tilde{\mu}^2\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\,\mathrm{LF}_{4,2,-1}[\tilde{\mu},m_2]\,-$$

$$2\,m_2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_2^4\,\mathrm{LF}_{5,1,-2}[\tilde{\mu},m_2]-6\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_1^2\,g_2^2\,(m_1+m_2)\,\mathrm{LF}_{2,2,1,-1}[m_2,\tilde{\mu},m_1]\,+$$

$$4\,m_1\,m_2\,g_1^2\,g_2^2\,\tilde{\mu}^2\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\,\mathrm{LF}_{2,2,1,0}[m_2,\tilde{\mu},m_1]\,+$$

$$4\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_1^2\,g_2^2\,(m_1+m_2)\,\mathrm{LF}_{3,2,1,-2}[m_2,\tilde{\mu},m_1]\,-$$

$$4\,m_1\,m_2\,g_1^2\,g_2^2\,\tilde{\mu}^2\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\,\mathrm{LF}_{3,2,1,-1}[m_2,\tilde{\mu},m_1]-\frac{1}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_e}^{pr}\,\overline{y_e}^{st}\,y_e^{sr}\,a_e^{pt}$$

$$\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s},m_{\tilde{e}}^{t}\right]-\frac{3}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_d}^{st}\,y_d^{sr}\,a_d^{pt}\,\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{t}\right]+$$

$$\frac{3}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}\,\left(\overline{y_d}^{pr}\,y_d^{sr}-\overline{y_u}^{pr}\,y_u^{sr}\right)\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}\right]+$$

$$\frac{3}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}\,\left(-\overline{y_d}^{pr}\,y_d^{sr}+\overline{y_u}^{pr}\,y_u^{sr}\right)\,\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}\right]+$$

$$\frac{3}{4}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}\,\left(-\overline{y_d}^{pr}\,y_d^{sr}+\overline{y_u}^{pr}\,y_u^{sr}\right)\,\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{q}}^{s}\right]+$$

$$\frac{3}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_u}^{st}\,y_u^{pt}\,a_d^{sr}\,\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}\right]+$$

$$\frac{3}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}\,\left(\overline{y_d}^{pr}\,y_d^{sr}-\overline{y_u}^{pr}\,y_u^{sr}\right)\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}\right]+$$

$$\frac{3}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}\,\left(-\overline{y_d}^{pr}\,y_d^{sr}+\overline{y_u}^{pr}\,y_u^{sr}\right)\,\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}\right]+$$

$$\frac{3}{4}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}\,\left(-\overline{y_d}^{pr}\,y_d^{sr}+\overline{y_u}^{pr}\,y_u^{sr}\right)\,\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p}\right]+$$

$$2\,m_1\,m_2\,g_1^2\,g_2^2\,\tilde{\mu}^2\,\mathrm{LF}_{2,1,1,0}[\tilde{\mu},m_1,m_2]-\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_1^2\,g_2^2\,(m_1+7\,m_2)\,\mathrm{LF}_{3,1,1,-1}[\tilde{\mu},m_1,m_2]\,+$$

$$2\,m_1\,m_2\,g_1^2\,g_2^2\,\tilde{\mu}^2\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\,\mathrm{LF}_{3,1,1,0}[\tilde{\mu},m_1,m_2]+6\,m_2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_1^2\,g_2^2\,\mathrm{LF}_{4,1,1,-2}[\tilde{\mu},m_1,m_2]\,-$$

$$3\,m_1\,m_2\,g_1^2\,g_2^2\,\tilde{\mu}^2\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\,\mathrm{LF}_{4,1,1,-1}[\tilde{\mu},m_1,m_2]+2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,g_1^2\,g_2^2\,(m_1+3\,m_2)$$

$$\mathrm{LF}_{3,2,1,-2}[\tilde{\mu},m_2,m_1]-4\,m_1\,m_2\,g_1^2\,g_2^2\,\tilde{\mu}^2\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\,\mathrm{LF}_{3,2,1,-1}[\tilde{\mu},m_2,m_1]\,-$$

$$\frac{3}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_d}^{st}\,\overline{a_d}^{pr}\,a_d^{pt}\,a_d^{sr}\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}\right]-$$

$$\frac{3}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}^3\,\overline{y_d}^{pr}\,\overline{y_d}^{st}\,y_d^{sr}\,a_d^{pt}\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}^{r},m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}\right]-$$

$$3\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}\,a_d^{pt}\,a_d^{sr}\,\mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{d}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}\right]+$$

$$\frac{3}{2}\,\tilde{\mu}\,\overline{y_d}^{st}\,a_d^{pt}\,a_d^{sr}\,\left(\overline{C_{\phi2\phi1}}\,\overline{a_d}^{pr}+\tilde{\mu}\,\overline{y_d}^{pr}\,\left(C_{\phi1^2}-C_{\phi2^2}\right)\right)\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}\right]+$$

$$\frac{3}{2}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}\,a_d^{pt}\,\left(a_d^{sr}\,\left(-C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_d^{sr}\right)\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}^{t},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}\right]-$$

$$\frac{1}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_e}^{st}\,\overline{a_e}^{pr}\,a_e^{pt}\,a_e^{sr}\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}\right]-$$

$$\frac{1}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}^3\,\overline{y_e}^{pr}\,\overline{y_e}^{st}\,y_e^{sr}\,a_e^{pt}\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}^{r},m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}\right]-$$

$$\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}\,a_e^{pt}\,a_e^{sr}\,\mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{e}}^{t},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}\right]+$$

$$\frac{1}{2}\,\tilde{\mu}\,\overline{y_e}^{st}\,a_e^{pt}\,a_e^{sr}\,\left(\overline{C_{\phi2\phi1}}\,\overline{a_e}^{pr}+\tilde{\mu}\,\overline{y_e}^{pr}\,\left(C_{\phi1^2}-C_{\phi2^2}\right)\right)\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}^{t},m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}\right]+$$

$$\frac{1}{2}\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}\,a_e^{pt}\,\left(a_e^{sr}\,\left(-C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_e^{sr}\right)\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}^{t},m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}\right]-$$

$$\tilde{\mu}\,\overline{y_e}^{st}\,a_e^{pt}\,a_e^{sr}\,\left(\overline{C_{\phi2\phi1}}\,\overline{a_e}^{pr}+\tilde{\mu}\,\overline{y_e}^{pr}\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\right)\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{t},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}\right]+$$

$$\tilde{\mu}\,\overline{y_e}^{st}\,a_e^{pt}\,a_e^{sr}\,\left(\overline{C_{\phi2\phi1}}\,\overline{a_e}^{pr}+\tilde{\mu}\,\overline{y_e}^{pr}\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\right)\,\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{t},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}\right]-$$

$$\overline{C_{\phi2\phi1}}\,\tilde{\mu}^3\,\overline{y_e}^{pr}\,\overline{y_e}^{st}\,y_e^{sr}\,a_e^{pt}\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}\right]+$$

$$\overline{C_{\phi2\phi1}}\,\tilde{\mu}^3\,\overline{y_e}^{pr}\,\overline{y_e}^{st}\,y_e^{sr}\,a_e^{pt}\,\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}\right]-$$

$$3\,\tilde{\mu}\,\overline{y_d}^{st}\,a_d^{pt}\,a_d^{sr}\,\left(\overline{C_{\phi2\phi1}}\,\overline{a_d}^{pr}+\tilde{\mu}\,\overline{y_d}^{pr}\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\right)\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}\right]+$$

$$3\,\tilde{\mu}\,\overline{y_d}^{st}\,a_d^{pt}\,a_d^{sr}\,\left(\overline{C_{\phi2\phi1}}\,\overline{a_d}^{pr}+\tilde{\mu}\,\overline{y_d}^{pr}\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\right)\,\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}\right]+$$

$$\frac{3}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}^3\,\overline{y_d}^{pr}\,y_d^{sr}\,\overline{y_u}^{st}\,a_u^{pt}\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}\right]+$$

$$\frac{3}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_d}^{st}\,a_d^{pt}\,\overline{a_u}^{pr}\,a_u^{sr}\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{u}}^{r}\right]-$$

$$3\,\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}\,a_u^{pt}\,a_u^{sr}\,\mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}\right]-\frac{3}{2}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}$$

$$\left(\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}\,a_u^{sr}+\tilde{\mu}\,\overline{y_u}^{pr}\,\left(2\,a_u^{sr}\,\left(C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_u^{sr}\right)\right)\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}\right]+$$

$$\frac{3}{2}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}\,\left(\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}\,a_u^{sr}+\tilde{\mu}\,\overline{y_u}^{pr}\,\left(2\,a_u^{sr}\,\left(C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_u^{sr}\right)\right)$$

$$\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}\right]+\frac{3}{4}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}$$

$$\left(-\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}\,a_u^{sr}+\tilde{\mu}\,\overline{y_u}^{pr}\,\left(4\,C_{\phi1^2}\,a_u^{sr}+3\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_u^{sr}\right)\right)\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}\right]+$$

$$\frac{3}{4}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}\,\left(\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}\,a_u^{sr}+\tilde{\mu}\,\overline{y_u}^{pr}\,\left(4\,C_{\phi2^2}\,a_u^{sr}+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_u^{sr}\right)\right)$$

$$\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}\right]-3\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}^3\,\overline{y_d}^{pr}\,\overline{y_d}^{st}\,y_d^{sr}\,a_d^{pt}\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}\right]+$$

$$3\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}^3\,\overline{y_d}^{pr}\,\overline{y_d}^{st}\,y_d^{sr}\,a_d^{pt}\,\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}\right]+\frac{3}{2}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_d^{sr}$$

$$\left(\overline{C_{\phi2\phi1}}\,\overline{a_d}^{pr}\,a_u^{pt}+\tilde{\mu}\,\overline{y_d}^{pr}\,\left(2\,a_u^{pt}\,\left(C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_u^{pt}\right)\right)\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}\right]-$$

$$\frac{3}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}\,\left(\overline{a_u}^{pr}\,a_u^{sr}+\tilde{\mu}^2\,\overline{y_u}^{pr}\,y_u^{sr}\right)\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}\right]+$$

$$\frac{3}{2}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}\,\left(\overline{a_u}^{pr}\,a_u^{sr}+\tilde{\mu}^2\,\overline{y_u}^{pr}\,y_u^{sr}\right)\,\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}\right]-$$

$$\frac{3}{4}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}\,\left(-3\,\overline{a_u}^{pr}\,a_u^{sr}+\tilde{\mu}^2\,\overline{y_u}^{pr}\,y_u^{sr}\right)\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}\right]+$$

$$\frac{3}{4}\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}\,\left(\overline{a_u}^{pr}\,a_u^{sr}+\tilde{\mu}^2\,\overline{y_u}^{pr}\,y_u^{sr}\right)\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}\right]-$$

$$3\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}\,\left(\overline{a_u}^{pr}\,a_u^{sr}+\tilde{\mu}^2\,\overline{y_u}^{pr}\,y_u^{sr}\right)\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}\right]+$$

$$3\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}\,\left(\overline{a_u}^{pr}\,a_u^{sr}+\tilde{\mu}^2\,\overline{y_u}^{pr}\,y_u^{sr}\right)\,\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}\right]-$$

$$3\,\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}\,a_u^{pt}\,a_u^{sr}\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}\right]+$$

$$3\,\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}\,a_u^{pt}\,a_u^{sr}\,\left(C_{\phi1^2}+C_{\phi2^2}\right)\,\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}\right]+\frac{3}{2}\,\tilde{\mu}\,\overline{y_u}^{st}\,a_u^{pt}$$

$$\left(\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}\,a_u^{sr}+\tilde{\mu}\,\overline{y_u}^{pr}\,\left(a_u^{sr}\,\left(C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_u^{sr}\right)\right)\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{u}}^{t},m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}\right]\Big)$$